Common-Source Amplifier Front-End

Analog Sensor Signal Conditioning Design and Simulation

Project Overview

Designed a Common-Source (CS) amplifier stage intended for use as an analog front-end (AFE) for small-signal sensor conditioning. The circuit was simulated in LTspice using TSMC 180nm CMOS parameters. To achieve high voltage gain, an active current mirror load was implemented instead of a standard resistive load, maximizing the output impedance.

Design Target vs. Simulated Results

Parameter	Target Specification	Simulated Result
Voltage Gain (A_V)	≥ 35 dB	38.2 dB
-3dB Bandwidth	≥ 50 MHz	62 MHz
Phase Margin	≥ 60°	65°
Power Dissipation	≤ 1.5 mW	1.1 mW

Simulation Methodologies

AC Analysis (Bode Plot)

A frequency sweep was conducted from 1 Hz to 1 GHz. The magnitude and phase responses were plotted to verify the closed-loop stability, gain bandwidth product (GBW), and to locate the dominant pole of the amplifier stage.

Transient Analysis

A 10 mV peak-to-peak sinusoidal input signal at 1 MHz was applied to verify large-signal behavior, output voltage swing, and to check for harmonic distortion or clipping at the supply rails (VDD = 1.8V).

VOLTAGE GAIN

38.2 dB

BANDWIDTH

62 MHz

V_{DD}

1.8 V